K. J. Somaiya School of Engineering, Mumbai-77

(Somaiya Vidyavihar University)

Department of Computer Engineering

Course Name:	Web Development Laboratory	**Semester:**	**III**
Date of Performance:	**__29_ / _07__ / ___2026___**	**DIV/ Batch No:**	**B-2**
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Experiment No: 1

Title

Design and Development of a Heatwave Intelligence Web Portal Front-End using HTML5

Aim of the Experiment

To design and develop a multi-page, use-case-based web portal front-end using HTML5 for a Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning system (KJS-CES-01), demonstrating proper document structure, semantic elements, formatting, lists, links, images, image maps, tables, forms, and media elements.

Objectives of the Experiment

1. Develop structured and semantic HTML5 documents for a real-world climate-intelligence dashboard.
2. Create interactive and media-rich web pages for heatwave monitoring and early warning.
3. Apply a real-world AI use case (heatwave monitoring) as the context for front-end web development.
4. Build a basic but complete front-end web portal that serves as the foundation for later CSS3, JavaScript, and React work.

COs to be achieved

CO1: Developing webpages using HTML and CSS.

Books / Journals / Websites references

1. *(Students should write — e.g., MDN Web Docs HTML5 reference, W3C.)*

Theory

HTML5 is the standard markup language used to structure content on the web. It introduces semantic elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer> that describe the meaning of content rather than only its appearance. HTML5 also

adds native support for media (<audio>, <video>), graphics (<canvas>, SVG), richer form controls (e.g., email, date, number, range input types with built-in validation), and embedding via <iframe>. These features make it well suited to building a decision-support dashboard such as a Heatwave Intelligence portal, where structured data (region-wise temperatures, forecasts, alerts) must be presented clearly and accessibly to meteorologists, disaster-management officers, and citizens.

Problem Statement

"Climate Intelligence Heatwave Monitoring & Early-Warning Web Portal"

A meteorological/disaster-management agency wants to develop a basic HTML5-based web portal front-end to display region-wise temperature data, heatwave forecasts, hotspot maps, alerts, and advisory services for meteorologists, disaster-management officers, and the public. Each task below contributes one page or feature of this portal, mirroring the components of the Heatwave Intelligence dashboard.

Task 1: Home Page – HTML5 Document Structure & Global Attributes

Use Case: Design the Home Page of the Heatwave Intelligence Portal.

Requirements:

- Use proper HTML5 document structure.
- Display the system name ("Heatwave Intelligence System"), a logo text, and the current monitoring season/year.
- Use global attributes (id, class, title, style).
- Add a tooltip on the system name explaining its purpose.

Deliverable: index.html

Skills Tested: Document structure, global attributes.

Task 2: Alert Description Page – Formatting & Text-Level Tags

Use Case: Create a page describing a Severe Heatwave Alert Bulletin.

Requirements:

- Highlight the alert name/region using and <mark>.
- Use for advisory themes and <small> for disclaimers (e.g., "forecast subject to revision").
- Use <sup> and <sub> for units and dates (e.g., temperature 47°C, bulletin version 2.1).
- Use <abbr> for technical terms (e.g., AWS, IMD, HW, SHW).

Deliverable: alert-details.html

Skills Tested: Formatting & text-level tags.

Code-

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Alert Details - Heatwave Intelligence System</title>
<style>
  body { margin: 40px; color: #000; }
  h1 { color: #2b0be2cc; }
  mark { background: #a80808; }
  small { display: block; margin-top: 20px; }
</style>
</head>
<body>
<h1>Heatwave Intelligence System</h1>
<p>Alert Bulletin Details</p>
<h2>Severe Heatwave Alert Bulletin</h2>
<p>
  <strong>Alert:</strong> <mark>Severe Heatwave Warning</mark> issued for
  <strong>Region:</strong> <mark>Vidarbha &amp;amp; Marathwada Zone</mark>
```

Maximum recorded temperature: 47^{°C}, exceeding normal by 6^{°C}.

Bulletin Version: 2_{.1}, issued by

<abbr title="Automatic Weather Station">AWS</abbr> network in coordination with the

<abbr title="India Meteorological Department">IMD</abbr>.

This bulletin classifies the current event as a

<abbr title="Heat Wave">HW</abbr>, escalating in select districts to a

<abbr title="Severe Heat Wave">SHW</abbr> condition. Citizens are advised to

avoid outdoor exposure between 12 PM and 4 PM, stay hydrated, and

follow local advisory updates issued by disaster-management authorities.

Vulnerable groups – children, the elderly, and outdoor workers – should take

extra precautionary measures during this period.

<small>

Disclaimer: This forecast is subject to revision as new data from

<abbr title="Automatic Weather Station">AWS</abbr> sensors and satellite feeds becomes available.

</small>

</body>

</html>

Output -



Task 3: About System Page – Block vs Inline Formatting

Use Case: Design an About the Heatwave Intelligence System page.

Requirements:

- Use sections for Objective, Data Sources, and Stakeholders.
- Use <blockquote> for a message from the Chief Meteorologist / Disaster Management Officer.
- Apply inline formatting inside paragraphs.
- Apply background colours to block sections (e.g., severity colour coding).

Deliverable: about-system.html

Skills Tested: Block-level and inline elements.

Code-

<!DOCTYPE html>

<html lang="en">

<head>

```
<meta charset="UTF-8">
<title>About System - Heatwave Intelligence System</title>
<style>
  body { margin: 40px; color: #3e0fe7; }
  h1 { color: #3c11da5d; }
  .objective { background: #fff8e1; padding: 10px 16px; }
  .data-sources { background: #e8f5e9; padding: 10px 16px; }
  .stakeholders { background: #ffebee; padding: 10px 16px; }
  blockquote { font-style: italic; margin: 20px 0; padding-left: 16px; border-left: 3px solid #999; }
</style>
</head>
<body>
<h1>Heatwave Intelligence System</h1>
<p>About the System</p>
<div class="objective">
  <h2>Objective</h2>
  <p>
    The system aims to provide <strong>early, reliable, and region-specific</strong>
    warnings of heatwave conditions, enabling <em>timely response</em> by
    meteorologists, disaster-management officers, and the general public.
  </p>
</div>
<div class="data-sources">
  <h2>Data Sources</h2>
  <p>
```

```
    Forecasts are generated using data from <strong>Automatic Weather Stations (AWS)</strong>,
    the <em>IMD historical database</em>, and real-time satellite feeds, combined through
    an AI-based prediction model for <strong>higher accuracy</strong>.
  </p>
</div>
<div class="stakeholders">
  <h2>Stakeholders</h2>
  <p>
    Key stakeholders include <strong>meteorologists</strong> who validate forecasts,
    <strong>disaster-management officers</strong> who coordinate ground response, and
    <em>citizens</em> who receive advisories to stay safe during severe heat events.
  </p>
</div>
<blockquote>
  "Timely and accurate heatwave intelligence saves lives. This system bridges the gap
  between raw weather data and actionable public advisories."
  <br>&mdash; Chief Meteorologist, Disaster Management Cell
</blockquote>
</body>
</html>
```

Output -



Task 4: Data Sources & Stakeholders Page – Lists & Nested Lists

Use Case: Display the Data Sources and Stakeholder Roles.

Requirements:

- Ordered list for the forecast/alert workflow steps.
- Unordered list for data sources (AWS sensors, IMD database, satellite feeds).
- Nested list showing stakeholders and the actions available under each role.
- Description list (<dl>) explaining abbreviations (AWS, IMD, HW, SHW).

Deliverable: sources.html

Skills Tested: List tags.

Task 5: Navigation Page – Hyperlinks

Use Case: Create a Navigation Menu for the portal.

Requirements:

- Link to all internal pages created so far.
- External link to the India Meteorological Department (IMD) website.
- Email link to the heatwave advisory coordinator.
- Telephone link for the disaster-management helpline.

Deliverable: navigation.html

Skills Tested: Hyperlink tags.

Task 6: Hotspot Map – Image & Image Map

Use Case: Provide an interactive Regional Heatwave Hotspot Map.

Requirements:

- Display a regional map image.
- Create an image map with clickable areas:
  - Region A (e.g., North Zone)
  - Region B (e.g., Central Zone)
- Each area opens a relevant region-detail page or link.

Deliverable: hotspot-map.html

Skills Tested: Image and image maps.

Task 7: Forecast Summary Page – Tables

Use Case: Display Region-wise Temperature & Forecast Results.

Requirements:

- Table showing Region, Station, Max Temperature, Severity Level.
- Use <caption>, <thead>, <tbody>, <tfoot>.
- Merge header cells using colspan (e.g., a "Forecast" header spanning multiple columns).

Deliverable: forecast.html

Skills Tested: Table tags.

Task 8: Advisory Subscription – Forms

Use Case: Create an Online Heatwave Advisory Subscription Form.

Requirements:

- Citizen name, email, mobile number.
- Region selection (dropdown).
- Preferred alert language (radio).
- Alert types to receive (checkbox).
- Submit and reset buttons.

Deliverable: subscribe.html

Skills Tested: Form tags.

Code –

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Subscribe - Heatwave Intelligence System</title>

<style>

  body { margin: 40px; color: #000; }

  h1 { color: #c5da0bde; }

  label { display: block; margin-top: 12px; }

  .inline-option { margin-right: 16px; }

</style>

</head>

<body>

<h1>Heatwave Intelligence System</h1>

<p>Heatwave Advisory Subscription</p>

<form action="#" method="post">

  <h2>Personal Details</h2>

  <label for="cname">Full Name</label>

  <input type="text" id="cname" name="cname" placeholder="Enter your full name"
required>

  <label for="cemail">Email Address</label>

  <input type="email" id="cemail" name="cemail" placeholder="you@example.com"
required>
```

```
  <label for="cmobile">Mobile Number</label>

  <input type="tel" id="cmobile" name="cmobile" pattern="[0-9]{10}" placeholder="10-digit
mobile number" required>

  <h2>Region</h2>

  <label for="region">Select your region</label>

  <select id="region" name="region" required>

    <option value="">-- Choose Region --</option>

    <option value="north">North Zone</option>

    <option value="central">Central Zone</option>

    <option value="konkan">Konkan Zone</option>

    <option value="vidarbha">Vidarbha Zone</option>

    <option value="marathwada">Marathwada Zone</option>

  </select>

  <h2>Preferred Alert Language</h2>

  <span class="inline-option"><input type="radio" id="lang-en" name="language"
value="english" checked> <label style="display:inline;" for="lang-
en">English</label></span>

  <span class="inline-option"><input type="radio" id="lang-hi" name="language"
value="hindi"> <label style="display:inline;" for="lang-hi">Hindi</label></span>

  <span class="inline-option"><input type="radio" id="lang-mr" name="language"
value="marathi"> <label style="display:inline;" for="lang-mr">Marathi</label></span>

  <h2>Alert Types to Receive</h2>

  <span class="inline-option"><input type="checkbox" id="alert-sms" name="alerttype"
value="sms" checked> <label style="display:inline;" for="alert-sms">SMS</label></span>

  <span class="inline-option"><input type="checkbox" id="alert-email" name="alerttype"
value="email"> <label style="display:inline;" for="alert-email">Email</label></span>
```

```
 <span class="inline-option"><input type="checkbox" id="alert-app" name="alerttype" value="app"> <label style="display:inline;" for="alert-app">App Notification</label></span>

 <p>

  <button type="submit">Subscribe</button>

  <button type="reset">Reset</button>

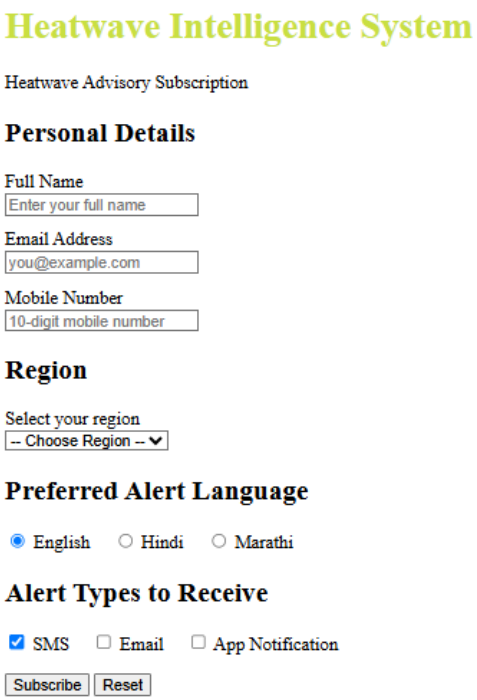
 </p>

</form>

</body>

</html>
```

Output-



Task 9: Media & Frames – Executable Content

Use Case: Create a Media & Awareness Page.

Requirements:

- Embed a previous heatwave awareness video using <video>.
- Add background audio advisory using <audio>.
- Embed a Google Map / IMD site using <iframe>.
- Use a <noscript> message for users with JavaScript disabled.

Deliverable: media.html

Skills Tested: Frames & executable tags.

Code :
Task – 2 Task – 3 Task-8 Put under the task

Output / Screenshot:
Attached under the task

Conclusion and Discussion:
Through this experiment, I gained hands-on experience in designing a multi-page front-end using HTML5 for a real-world use case — a Climate Intelligence Heatwave Monitoring and Early-Warning portal. I learned to structure content semantically and apply the appropriate tag for each type of information rather than relying on generic containers.